Panskura Banamali College (Autonomous)

B.Sc.(H) 3rd Semester Examination 2022 (Online mode)

Subject : BCA **Paper: GE3T**

Paper Name : Operation Research

F.M. : 60 **Time : 3 Hours**

Answer any four questions : **$4 \times 15 = 60$**

1. a) Solve graphically the LPP $Max\ Z = 2x_1 + 3x_2$

Subject to $x_1 + x_2 \leq 4,\ 6x_1 + 2x_2 \geq 8,\ x_1 + 5x_2 \geq 4,\ x_1 \leq 3,\ x_2 \leq 3,$ $x_1,\ x_2 \geq 0.$

b) A coin is to be minted containing 40% Silver and 50% Copper, 10% Nickle. The mint had available alloys A, B, C and D having the following compositions and costs.

	% Silver	% Copper	% Nickle	Costs
A	30	60	10	Rs. 11.00
B	35	35	30	Rs. 12.00
C	50	50	0	Rs. 16.00
D	40	45	15	Rs. 14.00

Present the problem of getting the alloys with specific composition at minimum cost in the form of LPP

c) Find graphically the feasible space of any for the following

$x_2 \geq 4,\ x_2 < 4$ and$x_1 \leq 4,\ x_1,\ x_2 \geq 0.$ 6+6+3

2. a)Find the dual of the following problem $Max\ Z = 5x_1 + 12x_2 + 4x_3$

Subject to $x_1 + 2x_2 + x_3 \leq 5,\ 2x_1 - x_2 + 3x_3 = 5,\ x_1 + x_2 - x_3 = 1,$ $x_1,\ x_2 \geq 0$and$x_3$is unrestricted

b) Solve the LPP $Max\ Z = 5x_1 + 11x_2$

Subject to $2x_1 + x_2 \leq 4,\ 3x_1 + 4x_2 \geq 24,\ 2x_1 - 3x_2 \geq 6,$ $x_1,\ x_2 \geq 0$ 7+8

3. a) Solve the following LPP using Big-M method $Max\ Z = 2x_1 - 3x_2$

Subject to $-x_1 + x_2 \geq -2,\ 5x_1 + 4x_2 \leq 46,\ 7x_1 + 2x_2 \geq 32,$

$x_1,\ x_2 \geq 0$

b) Solve the assignment problem

	I	*II*	*III*	*IV*
A	5	3	1	8
B	7	9	2	6
C	6	4	5	7
D	5	7	7	6

7+8

4. a) The following are the details of estimated times of activities of a certain project

Activity	*A*	*B*	*C*	*D*	*E*	*F*
Immediate Predecessor	----	*A*	*A*	*B*, *C*	----	*E*
Estimated Time (Weeks)	2	3	4	6	2	8

i) Construct a network of a project.

ii) Find the critical path and the expected time of the project.

iii) Calculate the earliest start time and earliest finish time for each activity.

iv) Calculate the float for each activity.

b) Distinguish between PERT and CPM. (4+4+2+2)+3

5. a) Determine the optimal sequence of jobs that minimizes the total elapsed time based on the following information (Processing time on machines is given in hours and passing is not allowed)

	Jobs						
	I	II	III	IV	V	VI	VII
Machine A_1	3	8	7	4	9	8	7
Machine A_2	4	3	2	5	1	4	3
Machine A_3	6	7	5	11	5	6	12

b) Define Total Elapsed time and Idle time.

c) In a balanced Transportation having m origins and n destinations $(m,\ n \geq 2)$,the exact number of basic variables is $m + n - 1$.

8+3+4

6. a) Solve using simplex method $Max\ Z = 3x_1 + x_2 + 3x_3$

Subject to $2x_1 + x_2 + x_3 \leq 2,\ x_1 + 2x_2 + 3x_3 \leq 5,\ 2x_1 + 2x_2 + x_3 \leq 6,$

$$x_1,\quad x_2,\quad x_3 \geq 0$$

b) Obtain the optimum basic feasible solution to the T.P.

	D_1	D_2	D_3	D_4	a_i
O_1	5	3	6	2	19
O_2	4	7	9	1	37
O_3	3	4	7	5	34
b_j	16	18	31	25	

7+8

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